TECHNICAL REPORT

LUN-TR-20261030-420D3A

LUNARIS

LUNAR UNCHARTED NAVIGATION, AUTONOMY & RESOURCE INTELLIGENCE SYSTEM

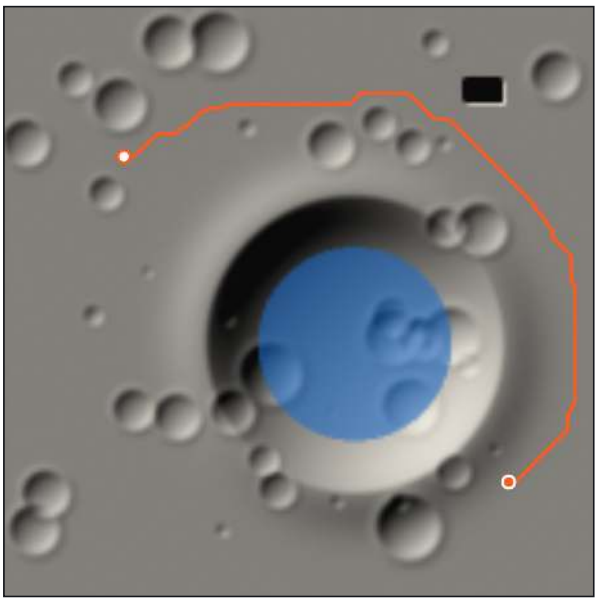
Traverse and Volatile-Prospect Assessment

SYNTHETIC demo terrain (not lunar data) · Lunar South Polar Region

Issued 30 October 2026 · Prepared with the LUNARIS analysis pipeline

DOCUMENT CONTROL

Report number	LUN-TR-20261030-420D3A
Date of issue	2026-10-30 09:35 UTC
Study region	SYNTHETIC demo terrain (not lunar data)
Dataset	SYNTHETIC demo terrain (not lunar data)
Region centre	—
Analysis grid	160 x 160 cells @ 80 m
Configuration	78c4f390d2ce6428
Dataset checksum	906fada484a5f499
Primary sources	NASA LRO: LOLA (PGDA), Diviner (PDS), LROC NAC; LOLA illumination models
Distribution	Unrestricted — derived from NASA open data



LOLA shaded relief, PSR (blue) and candidate traverse (orange)

EXECUTIVE SUMMARY

Summary of findings

This report documents an orbital-data assessment of the SYNTHETIC demo terrain (not lunar data) study area using the LUNARIS pipeline. Terrain derived from LOLA laser altimetry classifies 51% of the analysis area as low mobility hazard and 29% as impassable at the adopted slope and roughness thresholds. Permanently shadowed regions occupy 8.5% of the window. A hazard-weighted candidate traverse of 17.5 km was identified, with a maximum slope of 1.7° and no impassable cells crossed. The volatile investigation-priority model flags 8.2% of the area as candidates for further investigation. A traverse simulation of the NASA VIPER (Volatiles Investigating Polar Exploration Rover) returned **PASS**.

ROUTE LENGTH	MAX SLOPE ON ROUTE	PERMANENT SHADOW	ROVER SIMULATION
17.5 km	1.7°	8.5%	PASS

SECTION 01

Introduction and scope

LUNARIS is a decision-support system for permanently shadowed regions (PSRs) near the lunar poles. It combines laser altimetry, thermal-model products, illumination models and orbital imagery into traceable data layers, plans rover traverses over a documented cost surface, and evaluates them with a terramechanics-based rover simulation. Every layer in this report is classified as measured, derived, model estimate or illustrative, and cells without valid data are treated as unknown rather than safe.

Study region. Procedurally generated craters on a 12.8 km square. Does not describe any real location. Named features inside the window (IAU gazetteer): none.

SECTION 02

Data sources

Table 1. Input data products and checksums.

Layer	Product	Instrument	Units	SHA-256
elevation	demo_synthetic_dem_v1	none (synthetic)	m (synthetic)	5f198d0bd283
psr mask	demo_synthetic_psr_v1	none (synthetic)	1 = PSR (synthetic)	7e05b1f87689
thermal max	demo_synthetic_tmax_v1	none (synthetic)	K (synthetic)	12927bc3bce1

Table 2. Layers not ingested for this study region.

Layer	Status
ice stability depth	not provided by this dataset
image	not provided by this dataset
hydrogen	not provided by this dataset

Citation. Synthetic — generated by this repository

SECTION 03

Methods

Rasters are ingested with checksum verification, aligned to the elevation grid and block-averaged to the analysis resolution (cells require at least 75% valid sub-cells). Slope uses the Horn (1981) operator; roughness is the RMS residual from a local least-squares plane. Hazard classes take the worse of the slope and roughness classes; producer uncertainty raises a cell by one class. Traverses use an 8-connected A* search with an admissible octile heuristic over a per-metre cost multiplier.

Table 3. Processing record.

Stage	Detail
Ingest	elevation — synthetic_dem.tif
Ingest	psr mask — synthetic_psr.tif
Ingest	thermal max — synthetic_tmax.tif
Align	psr mask — identical grid (no resampling)
Align	thermal max — identical grid (no resampling)
Resample	factor 2 — nan-aware block mean
Analyse	0.02 s · configuration 78c4f390d2ce6428

SECTION 04

Terrain assessment

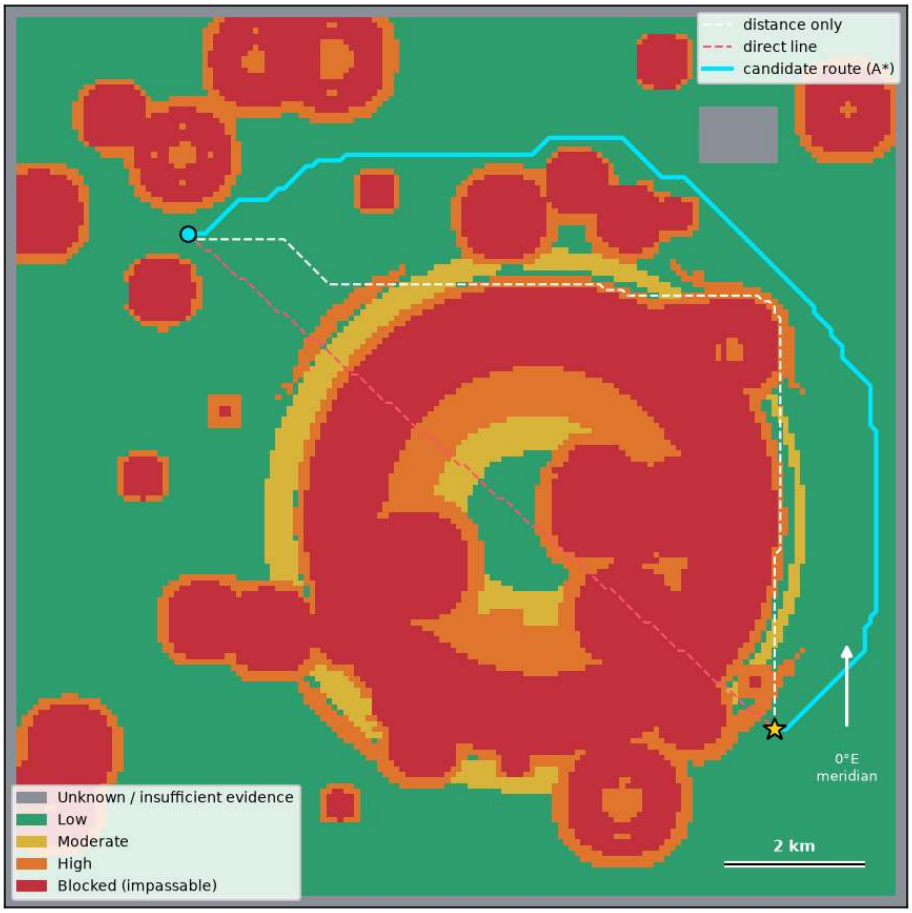


Figure 1. Mobility hazard classes with the candidate traverse (cyan), distance-only alternative and direct line. Up is the 0°E meridian direction of the polar stereographic grid.

Table 4. Hazard class distribution over the analysis area.

Hazard class	Area fraction
Unknown / insufficient evidence	5.5%
Low	50.5%
Moderate	3.6%
High	11.8%
Blocked (impassable)	28.6%

Boulder-scale hazards. grid is 80 m/px; rover-scale boulders (<10 m) are sub-pixel. Requires native-resolution NAC/ShadowCam imagery (~0.5–2 m/px) and a labelled training/evaluation set.

SECTION 05

Traverse planning

Table 5. Candidate traverse compared with rejected alternatives.

Metric	Candidate (hazard-weighted)	Distance-only	Direct line
Length (m)	17,492	14,817	11,236
Path cost	20,363	41,014	impassable
Maximum slope (°)	1.7	11.4	42.4
Mean slope (°)	0.8	6.6	12.2
Cells in PSR	0	0	40
Blocked cells crossed	0	0	34
Total ascent (m)	142	390	1,301

Path cost = sum over moves of step length x mean per-metre cost multiplier of the two cells; a value equal to the length means flat, certain, low-hazard terrain all the way. Relative units — not energy, time or risk probability.

Start 89.8370° S, 309.75° E; goal 89.8036° S, 130.64° E (planetocentric).

Table 6. Hazard exposure along the candidate traverse.

Hazard class along route	Cells
Unknown / insufficient evidence	0
Low	189
Moderate	0
High	0
Blocked (impassable)	0

SECTION 06

Volatile investigation priority

The priority model combines psr, thermal_max with explicit weights; layers not used: ice_stability_depth, hydrogen. A cell is assessed only when thermal evidence is present. Assessed fraction: 100.0%.

Table 7. Investigation-priority class distribution.

Priority class	Area fraction
Not assessed (insufficient evidence)	0.0%
Lower relative priority	91.6%
Moderate relative priority	0.1%
Candidate for further investigation	8.2%

Under 6 weight-perturbation scenarios, 0.3% of assessed cells change class at least once; the ranking should be read with this sensitivity in mind.

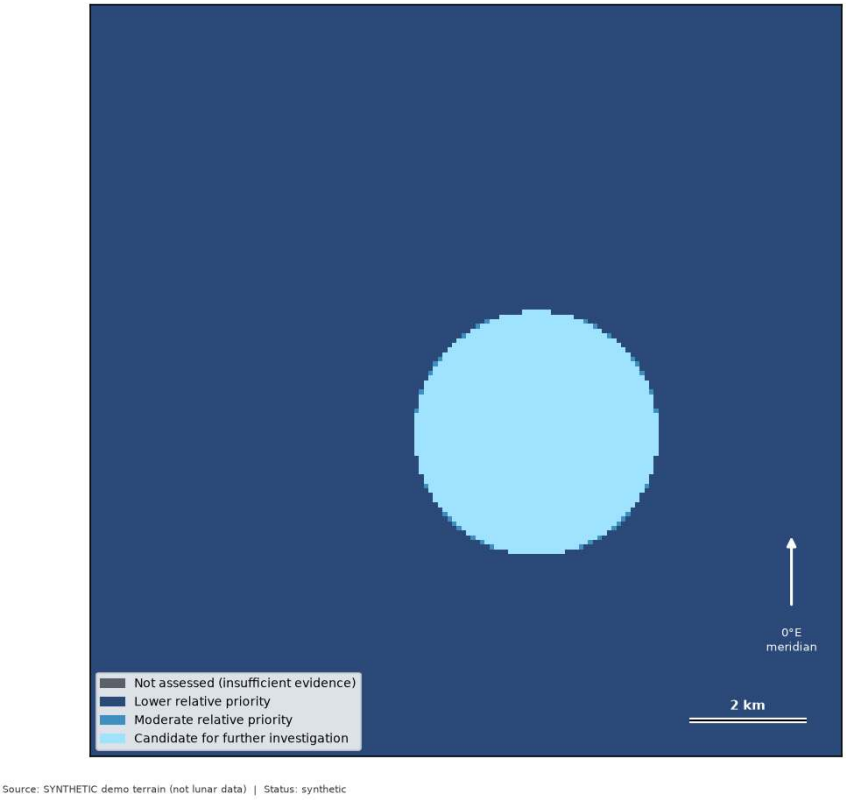


Figure 2. Relative investigation priority for water ice (ranking of evidence, not a detection).

SECTION 07

Rover traverse simulation

The NASA VIPER (Volatiles Investigating Polar Exploration Rover) was driven along the candidate traverse over the native-resolution elevation model on Lunar regolith — Apollo LRV design values, with Bekker–Wong wheel–soil forces, static stability and shadow-time checks. Verdict: **PASS**.

Table 8. Rover simulation results.

Quantity	Value
Distance driven / route length	17.49 / 17.49 km
Drive time (continuous)	53.5 h
Maximum slope / pitch / roll	5.4° / 5.4° / 4.1°
Wheel slip, maximum / mean	16% / 9%
Maximum sinkage	19.3 mm
Minimum static stability margin	28.3°
Time in PSR (longest continuous)	0.00 h (0.00 h)
Shadow energy / usable battery	0 / 4878 Wh
Loose-regolith sensitivity case	PASS — completed (17.49 km)

Unpublished rover parameters adopted as assumptions: ground clearance m, cg height m, drive efficiency, battery reserve frac. Terrain is the LOLA DEM at its native 20 m grid (bilinear). Rocks, boulders and craters smaller than ~20-40 m are invisible to this simulation; VIPER's published 15 cm obstacle limit cannot be checked from this data. Wheel-soil forces use the Bekker-Wong model with Apollo LRV soil parameters (Costes et al. 1972). Real PSR regolith properties are unmeasured and may differ; see the loose-soil sensitivity run. Quasi-static model: no dynamics, wheel-level suspension articulation, slip-sinkage or turn-in-place time.

SECTION 08

Assumptions

- Slope classes [10.0, 15.0, 25.0] deg and blocked >= 25.0 deg are prototype assumptions.
- Roughness high >= 2.0 m, blocked >= 6.0 m RMS (prototype).
- Cost weights {'slope': 8.0, 'roughness': 3.0, 'uncertainty': 4.0, 'psr': 1.5} are prototype choices; cost is relative, not energy or time.
- Ice-priority weights {'psr': 0.35, 'thermal_max': 0.4, 'ice_stability_depth': 0.25} with sensitivity +/-50%.
- Ground distance equals grid distance (polar stereographic scale error < 0.04 % within 2 deg of the pole).
- No-data and unknown cells are treated as blocked for routing.

SECTION 09

Limitations

- SYNTHETIC DATA: this report does not describe any real lunar location.
- Analysis grid 80.0 m/px: rover-scale obstacles (boulders, small craters) are not resolved.
- PSR map, temperatures and ice-stability depth are model products, each with its own epoch and resolution.
- No LEND hydrogen layer was used; no direct measurement of water ice is included.
- Illumination, communications line-of-sight and thermal survival are not modelled in the route.
- Layer 'Annual mean temperature, 2 cm depth (Diviner PRP)' unavailable: not provided
- Layer 'Ice stability depth (Diviner PRP)' unavailable: not provided by this dataset
- Layer 'LEND hydrogen' unavailable: not provided by this dataset

APPENDIX A

Layer catalogue

Table 9. Analysis layers, epistemic status and method.

Layer	Status	Units	Method
Elevation	synthetic	m	Block-mean 2x2 of source DEM (cells need >=75% valid sub-pixels).
Hillshade (visual aid)	illustrative	unitless	Lambertian shading of Horn gradients.
Slope	synthetic	degrees	Horn (1981) finite differences on the analysis grid; NaN propagates from no-data.
Roughness (detrended RMS)	synthetic	m	Least-squares plane removal per window (separable correlation).
Permanently shadowed regions	synthetic	fraction of cell	Block mean of the producer's 0/1 PSR map (illumination modelling on LOLA topography).
Annual max temperature (SYNTHETIC)	synthetic	K	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Annual mean temperature, 2 cm depth (Diviner PRP)	unavailable	-	-
Ice stability depth (Diviner PRP)	unavailable	-	-
LEND hydrogen	unavailable	-	-
Mobility hazard class	synthetic	class	Slope edges [10.0, 15.0, 25.0] deg, blocked >= 25.0 deg; roughness high >= 2.0 m, blocked >= 6.0 m.
Traversal cost multiplier	synthetic	cost per metre (>= 1)	See core/navigation/cost.py
Ice investigation priority (score)	synthetic	0–1 relative	core/ice_priority/evidence.py
Ice investigation priority (class)	synthetic	class	Thresholds at 1/3 and 2/3.
Evidence coverage	synthetic	fraction of weight	-
Priority sensitivity (std)	synthetic	score std	One-at-a-time perturbation.

Prepared with LUNARIS (Lunar Uncharted Navigation, Autonomy & Resource Intelligence System). Configuration 78c4f390d2ce6428; dataset checksum 906fada484a5f499; generated 2026-10-30T09:35:30 UTC.